

Component List: Smart Air Purifier System

1. Main Control & Interface

- **1x Arduino Nano (ATmega328P):** The brain of the system, responsible for reading sensor data and controlling the relay.
- **1x 16x2 I2C LCD Module:** Used for local real-time display of PM2.5 levels and fan status.
- **1x USB Mini-B Cable:** For power and sending data to the computer Serial Monitor.

2. Sensing & Actuation

- **1x MQ135 Gas Sensor Module:** Detects air pollutants (NH₃, NO_x, alcohol, benzene, smoke, CO₂).
- **1x 5V Single Channel Relay Module (Active LOW):** Acts as the switch for the high-power fan.
- **1x DC Fan:** The physical air purification component (5V or 12V depending on your external power source).

3. Power & Connectivity

- **1x External Power Source:** A battery or DC adapter suitable for your fan's voltage rating (e.g., 9V battery or 12V adapter).
- **Jumper Wires (Male-to-Female & Male-to-Male):** For connecting modules to the Nano.
- **1x Breadboard (Optional):** For easy prototyping and sharing 5V/GND rails.